Houlin (Richard) Wang

(310) 500-7579 | hwang374@ucsc.edu | Santa Cruz, CA
linkedin.com/in/houlinwang | houlin00.github.io

Computational linguist and speech scientist with 5+ years of experience in ASR evaluation, forced alignment, and corpus analysis for low-resource and non-standard speech

Education

Doctor of Philosophy (Ph.D.) in Linguistics Expected June 2027
University of California, Santa Cruz Santa Cruz, CA
Specialization in Speech Technology & Computation

Master of Arts (M.A.) in Linguistics, GPA 3.97 June 2022
University of California, Los Angeles Los Angeles, CA

Bachelor of Arts (B.A.) in Linguistics, GPA 3.95, Magna Cum Laude June 2022
Specialization in Computation • Japanese minor • Department Scholar and Departmental Honors

Experience

Computational Linguistics Researcher September 2022 – Present
University of California, Santa Cruz Santa Cruz, CA

- Build Python and R pipelines for acoustic feature extraction, automated annotation, and quality control, analyzing speech at 10x the scale of manual methods
- Run mixed-effects regression and ML classification across 20,000+ model fits and 8 variables, showing duration does not predict consonant lenition in Mandarin
- Develop a language-agnostic lenition detector for corpus quality monitoring, trained on public Mandarin corpora and tested on my own recordings of 30 speakers (20+ hours)
- Design online speech-perception experiments (80+ participants) and model listener decision boundaries in R

AI Evaluation Researcher June 2026 – Present
Living Tongues Institute for Endangered Languages Remote

- Evaluate Meta's Omnilingual ASR on 5 low-resource Munda languages, comparing WER, CER, and phone matching on trained and zero-shot languages
- Diagnose decoder looping, wrong-script output, and mislabeled training data, and write evaluation reports with recommendations
- Annotate 90+ hours of speech audio and standardize quality control across a remote team

Research Mentor & Instructor June 2026 – August 2026
Science Internship Program, University of California, Santa Cruz Santa Cruz, CA

- Designed and taught an 8-week curriculum in Python, R, Praat, and hypothesis testing
- Mentored 3 student researchers in annotating and modeling 2,500+ speech tokens

Computational Linguistics Research Assistant September 2019 – June 2021
UCLA Computational Phonology & Language Processing Labs Los Angeles, CA

- Built Python algorithms to analyze phonotactic patterns across 10,000+ lexicon entries
- Created reproducible R pipelines for corpus analysis and coordinated protocols for 30+ study participants

Skills

Speech & NLP: ASR evaluation (WER, CER), forced alignment (MFA), Praat, corpus annotation
ML & Statistics: Mixed-effects regression (lme4), ML classification, bootstrapping, hypothesis testing
Programming: Python (pandas, NumPy, SciPy), R (tidyverse), SQL, PyTorch, Hugging Face, Git, Bash, LaTeX
Languages: Mandarin (native), English (fluent), German and Japanese (working)

Publications

Wang, R. (2025). Distribution of tone affects prosodic parsing in Beijing Mandarin. *Proceedings of AMP, 1*(1)
Wang, R. (2025). Morphosyntax-prosody mismatch in Beijing Mandarin. *Proceedings of WCCFL 41*, 592–599
Wang, R. et al. (under review). Does Omnilingual work? Testing ASR on Munda languages. *ComputEL-10*

Awards

THI Summer Fellowship ($6,000) • THI Dissertation Fellowship ($5,000) • SIP Fellowship ($4,750), all 2026